

JKBOSE 2026

CLASS 11TH

**LIMITS AND
DERIVATIVES**

MATHS

LECTURE-4

BY-MEHRAJ SIR



TOPICS TO BE COVERED

1

Limits of Trigonometric Functions

2

Exercise Discussion 12.1

3

Related Problems

Theorem – 1:

Let f and g be two functions such that both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist.

Then

(i) Limit of sum of two functions is sum of the limits of the functions, i.e.,

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

(ii) Limit of difference of two functions is difference of the limits of the function, i.e.,

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

(iii) Limit of product of two functions is product of the limits of the functions, i.e.,

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

(iv) Limit of quotient of two functions is quotient of the limits of the functions (whenever the denominator is non zero), i.e.,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

Theorem – 2:

For any positive integer n , $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$.

$$\begin{aligned}
 * \quad \lim_{x \rightarrow 5} \left(\frac{x^2 - 25}{x - 5} \right) &= \lim_{x \rightarrow 5} \left(\frac{x^2 - 5^2}{x - 5} \right) = 2 \times 5^{2-1} \\
 &= 2 \times 5 \\
 &= 10.
 \end{aligned}$$

EXAMPLE

Find the positive integer n so that $\lim_{x \rightarrow 3} \frac{x^n - 3^n}{x - 3} = 108$.

(Sol) Given that $\lim_{x \rightarrow 3} \left(\frac{x^n - 3^n}{x - 3} \right) = 108$.

$$\Rightarrow n \times 3^{n-1} = 108 \quad \left(\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1} \right)$$

$$\begin{array}{r|l} 3 & 108 \\ \hline 3 & 36 \\ \hline 3 & 12 \\ \hline & 4 \end{array}$$

$$\Rightarrow n \times 3^{n-1} = 4 \times 3^3$$

$$\Rightarrow n \times 3^{n-1} = 4 \times 3^{4-1} \Rightarrow \boxed{n=4} \text{ Hence the result.}$$

QUESTION (NCERT EXAMPLER):

$$\lim_{x \rightarrow 1} \frac{x^m - 1}{x^n - 1} \text{ is}$$

1 1 ✗

2 $\frac{m}{n}$ ✓

3 $-\frac{m}{n}$ ✗

4 $\frac{m^2}{n^2}$ ✗

(Sol) We have $\lim_{x \rightarrow 1} \left(\frac{x^m - 1}{x^n - 1} \right) = \lim_{x \rightarrow 1} \left[\frac{\left(\frac{x^m - 1}{x - 1} \right)}{\left(\frac{x^n - 1}{x - 1} \right)} \right]$

$$\lim_{x \rightarrow a} \left(\frac{x^n - a^n}{x - a} \right) = na^{n-1}$$

$$= \frac{\lim_{x \rightarrow 1} \left(\frac{x^m - 1}{x - 1} \right)}{\lim_{x \rightarrow 1} \left(\frac{x^n - 1}{x - 1} \right)} = \frac{m(1)^{m-1}}{n(1)^{n-1}} = \frac{m}{n}$$

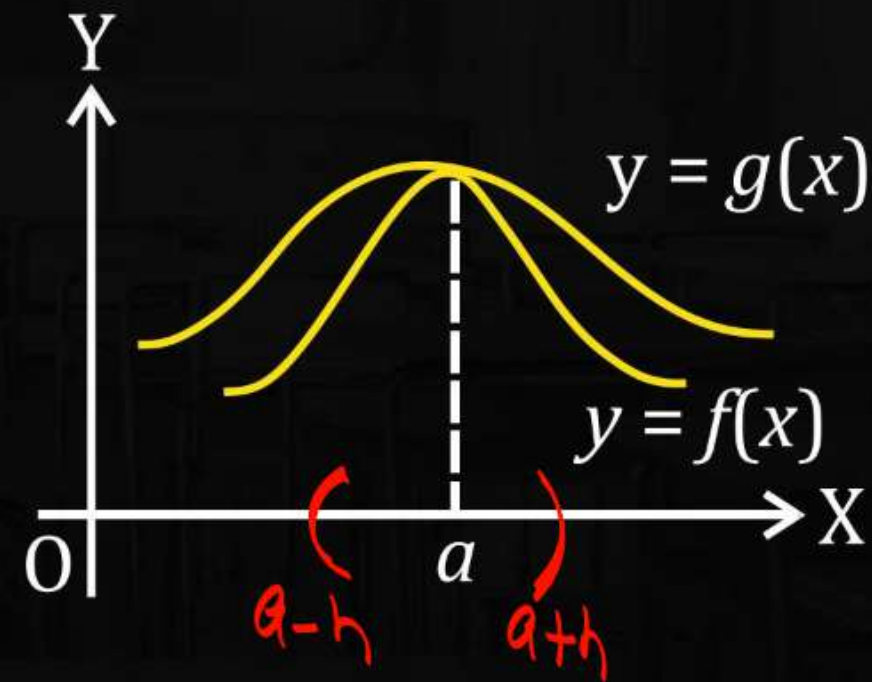
Theorem – 3:

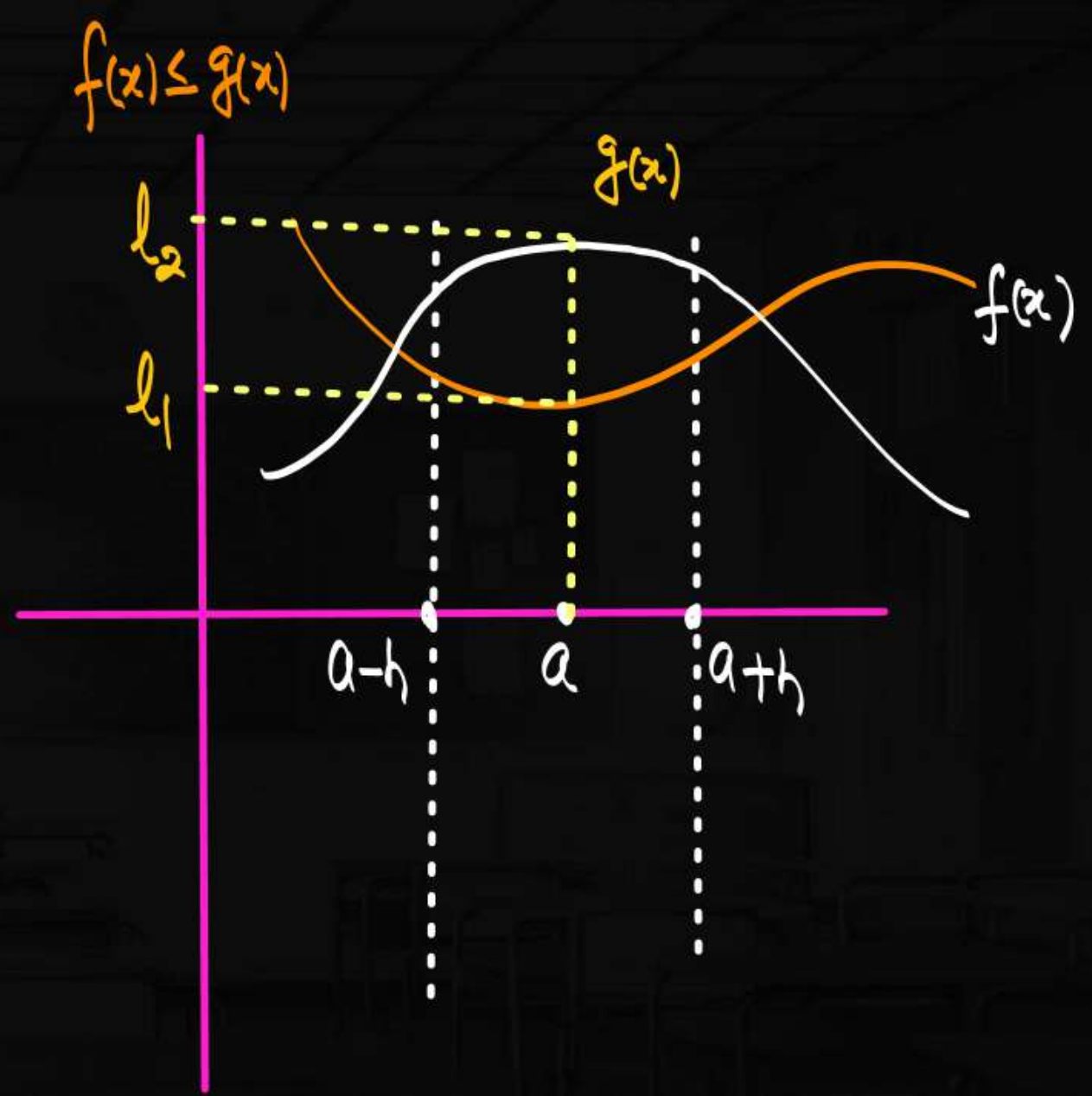
Let f and g be two real valued functions with the same domain such that $f(x) \leq g(x)$ for all x in the domain of definition, For some a , if both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, then $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$. This is illustrated in Figure.

$$* f(x) \leq g(x) \quad \forall x \in D.$$

$$* \lim_{x \rightarrow a} f(x) \text{ \& \& } \lim_{x \rightarrow a} g(x) \text{ exists}$$

$$\Rightarrow \lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x).$$





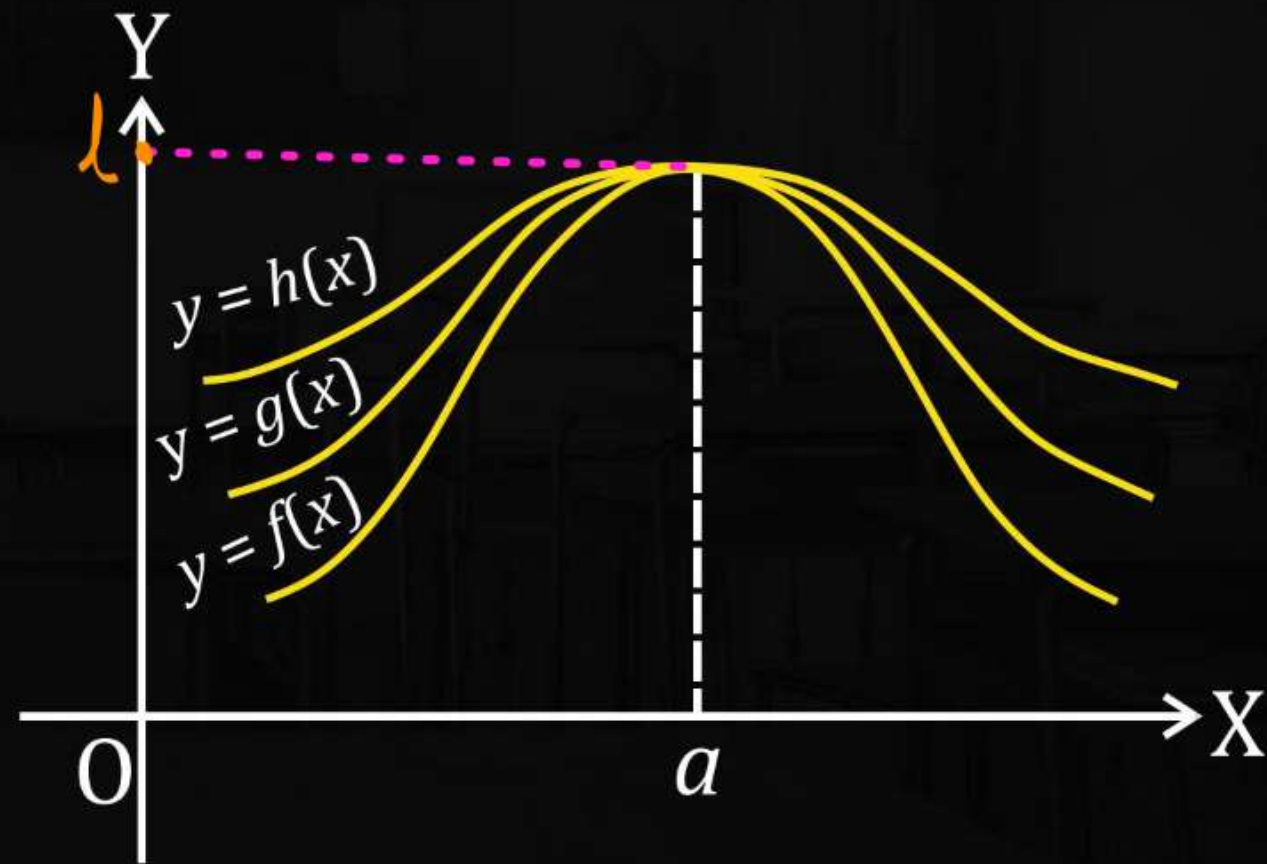
Theorem – 4: (Sandwich Theorem / Squeeze principle)

Let f, g and h be real functions such that $f(x) \leq g(x) \leq h(x)$ for all x in the common domain of definition. For some real number a , if $\lim_{x \rightarrow a} f(x) = l = \lim_{x \rightarrow a} h(x)$, and $\lim_{x \rightarrow a} g(x) = l$. This is illustrated in Figure.

$$* f(x) \leq g(x) \leq h(x) \quad \forall x \in D.$$

$$* \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = l.$$

$$\Rightarrow \lim_{x \rightarrow a} g(x) = l.$$



Limits of Trigonometric Functions

Theorem – 5:

The following are two important limits.

$$(i) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1. \star$$

$$(ii) \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0 \star$$

$$(iii) \quad \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right) = 1. \star$$

QUESTION (NCERT EXAMPLER):

$\lim_{x \rightarrow 0} \frac{\sin x}{x(1 + \cos x)}$ is equal to

1 0 ✗

2 $1/2$ ✓

3 1 ✗

4 -1 ✗

(Sol) Here $\lim_{x \rightarrow 0} \frac{\sin x}{x(1 + \cos x)} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \times \frac{1}{1 + \cos x} \right)$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \times \lim_{x \rightarrow 0} \left(\frac{1}{1 + \cos x} \right)$$

$$= 1 \times \frac{1}{1 + \cos 0} = \frac{1}{1 + 1} = \frac{1}{2}$$

QUESTION (NCERT EXAMPLER):

$\lim_{x \rightarrow \pi/2} \frac{1 - \sin x}{\cos x}$ is equal to

1 0 ✓

2 -1 ✗

3 1 ✗

4 ✗ Does not exist

(Sol) Here $\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\cos x}$ | $\frac{0}{0}$ form.

$$= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{(1 - \sin x)(1 + \sin x)}{(1 + \sin x) \cos x} \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 x}{(1 + \sin x) \cos x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cancel{\cos^2 x}}{(1 + \sin x) \cancel{\cos x}}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\cos x}{1 + \sin x} \right) = \frac{\cos(\frac{\pi}{2})}{1 + \sin(\frac{\pi}{2})}$$

$$= \frac{0}{2} = 0.$$

QUESTION (NCERT EXAMPLER):

$\lim_{x \rightarrow 0} x \sin \frac{1}{x}$ is equal to

1 0 ✓

2 1 ✗

3 $1/2$ ✗

4 Does not exist ✗

$$\begin{aligned} & \lim_{x \rightarrow 0} \left(x \sin \frac{1}{x} \right) \\ &= 0 \times \sin(\infty) \\ &= 0 \times (\text{finite no. b/w } -1 \text{ \& } 1) \\ &= 0. \end{aligned}$$

$$\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$$

EXERCISE – 12.1

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} (x + 3) = 0 + 3 = 3.$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow \pi} \left(x - \frac{22}{7} \right) = \pi - \frac{22}{7}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 1} \pi x^2 = \pi(1)^2 = \pi$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 4} \frac{4x + 3}{x - 2} = \frac{4(4) + 3}{4 - 2} = \frac{19}{2}.$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow -1} \frac{x^{10} + x^5 + 1}{x - 1} = \frac{(-1)^{10} + (-1)^5 + 1}{-1 - 1}$$

$$= \frac{1 - \cancel{1} + \cancel{1}}{-2} = -\frac{1}{2}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} \frac{(x+1)^5 - 1}{x}$$

(Sol) let $y = \lim_{x \rightarrow 0} \frac{(x+1)^5 - 1}{x}$ $\left| \frac{0}{0} \text{ form} \right|$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{({}_5C_0 x^5 + {}_5C_1 x^4 + {}_5C_2 x^3 + {}_5C_3 x^2 + {}_5C_4 x^1 + {}_5C_5) - 1}{x}$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{(x^5 + 5x^4 + 10x^3 + 10x^2 + 5x + 1) - 1}{x}$$

$$\begin{aligned} (x+1)^5 &= {}_5C_0 x^5 + {}_5C_1 x^4(1) + {}_5C_2 x^3(1)^2 + {}_5C_3 x^2(1)^3 \\ &\quad + {}_5C_4 x(1)^4 + {}_5C_5(1)^5 \end{aligned}$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{x^5 + 5x^4 + 10x^3 + 10x^2 + 5x + \cancel{1} - \cancel{1}}{x}$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{x(x^4 + 5x^3 + 10x^2 + 10x + 5)}{x}$$

$$\Rightarrow y = \lim_{x \rightarrow 0} (x^4 + 5x^3 + 10x^2 + 10x + 5)$$

$$\Rightarrow \boxed{y = 5} \text{ Hence the result.}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 2} \frac{3x^2 - x - 10}{x^2 - 4}$$

(Sol) let $y = \lim_{x \rightarrow 2} \left(\frac{3x^2 - x - 10}{x^2 - 4} \right) \Bigg| \frac{0}{0} \text{ form}$

$$\Rightarrow y = \lim_{x \rightarrow 2} \left(\frac{3x^2 - 6x + 5x - 10}{x^2 - 2^2} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow 2} \left(\frac{3x(x-2) + 5(x-2)}{(x+2)(x-2)} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}(3x+5)}{(x+2)\cancel{(x-2)}}$$

$$\Rightarrow y = \lim_{x \rightarrow 2} \left(\frac{3x+5}{x+2} \right) = \frac{3 \times 2 + 5}{2 + 2}$$

$$\Rightarrow y = \frac{11}{4} \text{ Hence the result.}$$

QUESTION

(PYQ JKBOSE)

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 3} \frac{x^4 - 81}{2x^2 - 5x - 3}$$

(Sol) let $y = \lim_{x \rightarrow 3} \frac{x^4 - 81}{2x^2 - 5x - 3}$ | $\frac{0}{0}$ form.

$$\Rightarrow y = \lim_{x \rightarrow 3} \frac{(x^2)^2 - 9^2}{2x^2 - 6x + x - 3}$$

$$\Rightarrow y = \lim_{x \rightarrow 3} \frac{(x^2 + 9)(x^2 - 9)}{2x(x-3) + 1(x-3)}$$

$$\Rightarrow y = \lim_{x \rightarrow 3} \frac{(x^2 + 9)(x^2 - 3^2)}{(2x + 1)(x - 3)}$$

$$\Rightarrow y = \lim_{x \rightarrow 3} \frac{(x^2 + 9)(x + 3)(x - 3)}{(2x + 1)(x - 3)}$$

$$\Rightarrow y = \lim_{x \rightarrow 3} \frac{(x^2 + 9)(x + 3)}{(2x + 1)} = \frac{(9 + 9)(3 + 3)}{(2 \times 3 + 1)}$$

$$\Rightarrow y = \frac{18 \times 6}{7}$$

$$\Rightarrow \boxed{y = \frac{108}{7}}$$

Hence the result.

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} \frac{ax + b}{cx + 1} \quad \text{H/W}$$

QUESTION

$$z^{\frac{1}{3}} = (z^{\frac{1}{6}})^2$$

Evaluate the following limits in Exercises 1 to 22.

(PYQ JKBOSE)

$$\lim_{z \rightarrow 1} \frac{z^{\frac{1}{3}} - 1}{z^{\frac{1}{6}} - 1}$$

(sol) let $y = \lim_{z \rightarrow 1} \frac{z^{\frac{1}{3}} - 1}{z^{\frac{1}{6}} - 1}$ $\left| \frac{0}{0} \text{ form} \right|$

$$\Rightarrow y = \lim_{z \rightarrow 1} \frac{(z^{\frac{1}{6}})^2 - 1^2}{(z^{\frac{1}{6}} - 1)} = \lim_{z \rightarrow 1} \frac{(z^{\frac{1}{6}} + 1)(\cancel{z^{\frac{1}{6}} - 1})}{(\cancel{z^{\frac{1}{6}} - 1})} = (1)^{\frac{1}{6}} + 1 = 2$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 1} \frac{ax^2 + bx + c}{cx^2 + bx + a}, \quad a + b + c \neq 0$$

H/W

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 1} \frac{\frac{1}{x} + \frac{1}{2}}{x + 2} \quad \text{H/W}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 1} \frac{\sin ax}{bx} \quad \text{H/W}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}, a, b \neq 0$$

(Sol) let $y = \lim_{x \rightarrow 0} \left(\frac{\sin ax}{\sin bx} \right)$ $\left| \frac{0}{0} \text{ form} \right.$

$$\Rightarrow y = \lim_{x \rightarrow 0} \left\{ \frac{\frac{\sin ax}{ax} \times a}{\frac{\sin bx}{bx} \times b} \right\}$$

$$\Rightarrow y = \frac{a}{b} \lim_{x \rightarrow 0} \left\{ \frac{\frac{\sin ax}{ax}}{\frac{\sin bx}{bx}} \right\}$$

$$\Rightarrow y = \frac{a}{b} \left\{ \frac{\lim_{x \rightarrow 0} \frac{\sin ax}{ax}}{\lim_{x \rightarrow 0} \frac{\sin bx}{bx}} \right\} = \frac{a}{b} \left(\frac{1}{1} \right) = \frac{a}{b}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow \pi} \frac{\sin(\pi - x)}{\pi(\pi - x)}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow \pi^+} \frac{\sin(\pi - x)}{\pi(\pi - x)} = \lim_{h \rightarrow 0} \frac{\sin[\pi - (\pi + h)]}{\pi(\pi - (\pi + h))} \\
 &= \frac{1}{\pi} \lim_{h \rightarrow 0} \frac{\sin(\cancel{\pi - \pi} - h)}{(\cancel{\pi - \pi} - h)} = \frac{1}{\pi} \lim_{h \rightarrow 0} \frac{\sin(-h)}{-h} \\
 &\quad \left(\frac{1}{\pi} \times \lim_{h \rightarrow 0} \frac{\sin h}{h} \right) \\
 &= \frac{1}{\pi} \times 1 \\
 &= \frac{1}{\pi}
 \end{aligned}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} \frac{\cos x}{\pi - x} \quad \text{H/W}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} \frac{\cos 2x - 1}{\cos x - 1} \quad (\text{Important})$$

(Sol) let $y = \lim_{x \rightarrow 0} \frac{\cos 2x - 1}{\cos x - 1} \mid \frac{0}{0} \text{ form}$

$$\Rightarrow y = \lim_{x \rightarrow 0} \left(\frac{2 \cos^2 x - 1 - 1}{\cos x - 1} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \left(\frac{2 \cos^2 x - 2}{\cos x - 1} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{2(\cos^2 x - 1)}{(\cos x - 1)}$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{2(\cos x + 1)(\cancel{\cos x - 1})}{(\cancel{\cos x - 1})}$$

$$\Rightarrow y = \lim_{x \rightarrow 0} 2(\cos x + 1)$$

$$\Rightarrow y = 2(1 + 1) = 4$$

$$\Rightarrow \boxed{y = 4}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} \frac{ax + x \cos x}{b \sin x}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{x(a + \cos x)}{b \sin x} = \lim_{x \rightarrow 0} \frac{(a + \cos x)}{b \left(\frac{\sin x}{x} \right)} \\
 &= \frac{\lim_{x \rightarrow 0} (a + \cos x)}{b \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)} = \frac{a+1}{b \times 1} = \frac{a+1}{b}
 \end{aligned}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} x \sec x$$

h/w

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} \frac{\sin ax + bx}{ax + \sin bx} \quad a, b, a + b \neq 0,$$

h/w

QUESTION

Evaluate the following limits in Exercises 1 to 22.

(Important)

$$\lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x)$$

(Sol) let $y = \lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x)$ | $\infty - \infty$ form

$$\Rightarrow y = \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{\cos x}{\sin x} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{\sin x} \right) \left| \frac{0}{0} \text{ form} \right.$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \left(\frac{(1 - \cos x)(1 + \cos x)}{\sin x (1 + \cos x)} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{\sin x (1 + \cos x)}$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{\sin^2 x}{\cancel{\sin x} (1 + \cos x)}$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{\sin x}{1 + \cos x} = \frac{\sin 0}{1 + \cos 0} = \frac{0}{1+1} = \frac{0}{2} = 0$$

$$\Rightarrow \boxed{y = 0}$$

QUESTION

Suppose $f(x) = \begin{cases} a + bx, & x < 1 \\ 4, & x = 1 \\ b - ax, & x > 1 \end{cases}$

and if $\lim_{x \rightarrow 1} f(x) = f(1)$ what are possible values of a and b ?

(Sol) Given that $\lim_{x \rightarrow 1} f(x) = f(1) = 4$.

$$\Rightarrow \lim_{x \rightarrow 1} f(x) = 4$$

Therefore $\lim_{x \rightarrow 1^+} f(x) = 4$ & $\lim_{x \rightarrow 1^-} f(x) = 4$.

$$\Rightarrow \lim_{h \rightarrow 0} f(1+h) = 4 \text{ & } \lim_{h \rightarrow 0} f(1-h) = 4$$

$$\Rightarrow \lim_{h \rightarrow 0} (b - a(1+h)) = 4 \text{ & } \lim_{h \rightarrow 0} (a + b(1-h)) = 4$$

$$\Rightarrow b - a = 4 \text{ --- (1) & } a + b = 4 \text{ --- (2)}$$

Adding (1) & (2), we get;

$$b - a + a + b = 8$$

$$\Rightarrow 2b = 8$$

$$\Rightarrow b = 4$$

$$\text{So, } a + 4 = 4$$

$$a = 0$$

Therefore $a = 0, b = 4$.

Hence the result.

QUESTION

Let $a_1, a_2, \dots, a_n$ be fixed real numbers and define a function

$$f(x) = (x - a_1)(x - a_2) \dots (x - a_n)$$

Where is $\lim_{x \rightarrow a_1} f(x)$? For some $a \neq a_1, a_2, \dots, a_n$, compute $\lim_{x \rightarrow a} f(x)$

H/W

Marifat Yousuf



Tamanna



Jan-Mehraj 1729.
forqanx

Thank
You